

CFB PETAWAWA, ON, Canada

WMO: 716250

Lat: 45.948N

Lon: 77.311W

Elev: 427

StdP: 14.47

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-13.1	-28.1	1.2	-18.1	-22.6	1.6	-12.0	21.0	12.9	18.9	13.3	2.8	280	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	87.4	70.8	84.0	68.9	80.8	67.7	73.6	83.3	71.7	80.2	69.9	77.2	8.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.6	114.5	77.7	68.9	107.9	75.8	67.2	101.5	74.0	37.5	83.3	35.7	80.1	34.2	77.2	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	16.1	14.1	-27.4	92.8	5.4	3.1	-31.3	95.0	-34.4	96.8	-37.4	98.6	-41.4	100.8		
			-27.7	76.9	5.3	1.9	-31.5	78.3	-34.6	79.4	-37.6	80.4	-41.4	81.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	11.5	15.1	25.9	40.6	54.0	63.2	67.7	65.8	58.2	46.0	33.1	19.9	
	DBStd	21.73	13.80	12.10	11.48	8.45	8.08	6.88	5.56	5.73	7.73	8.36	9.40	12.36	
	HDD50	4913	1195	978	750	302	48	1	0	0	15	181	510	933	
	HDD65	8739	1660	1398	1213	732	352	114	33	60	231	590	958	1398	
	CDD50	1953	0	0	1	20	173	397	549	491	262	58	2	0	
	CDD65	304	0	0	0	0	12	61	117	86	27	1	0	0	
	CDH74	2992	0	0	1	22	204	620	1144	757	228	16	0	0	
	CDH80	851	0	0	0	5	52	188	350	204	50	2	0	0	
Wind	WSAvg	5.9	6.1	6.2	6.8	6.8	6.0	5.3	4.9	4.8	5.1	6.2	6.4	6.3	
Precipitation	PrecAvg	33.5	2.1	1.8	2.6	2.6	2.8	3.5	2.9	3.3	3.2	2.9	2.8	3.0	
	PrecMax	45.4	4.0	4.4	4.6	4.0	5.3	8.8	5.6	6.5	6.5	5.6	4.5	6.1	
	PrecMin	26.9	0.7	0.2	1.1	0.6	0.7	1.7	1.4	1.0	1.7	1.3	1.3	1.2	
	PrecStd	4.4	0.7	1.0	0.9	1.0	1.4	1.7	1.1	1.7	1.2	1.2	1.0	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.5	44.8	58.3	77.0	85.7	90.0	92.2	90.9	85.8	75.4	60.0	48.7	
		MCWB	43.3	39.3	46.7	60.2	65.9	71.1	74.1	72.6	72.6	62.4	53.6	45.3	
	2%	DB	38.7	38.9	50.2	67.5	79.7	85.5	87.6	85.3	79.9	68.6	55.0	42.7	
		MCWB	36.8	35.5	42.7	52.3	62.4	69.6	71.2	69.9	68.7	59.9	50.5	40.7	
	5%	DB	35.1	36.1	45.1	60.9	74.5	81.1	83.9	81.5	75.9	63.8	50.2	38.2	
		MCWB	33.2	33.3	38.8	49.0	59.9	67.0	68.9	68.6	66.2	57.2	46.5	36.2	
	10%	DB	31.8	33.2	41.1	55.3	69.8	76.9	80.4	78.1	71.9	59.5	46.1	35.4	
		MCWB	30.4	31.3	36.4	44.7	57.7	64.9	68.1	66.9	64.2	54.2	42.8	33.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	40.1	51.5	61.7	69.9	74.6	76.8	74.9	73.9	65.9	55.5	45.8
			MCDB	44.0	42.5	56.6	72.3	80.9	85.5	87.7	85.2	82.4	71.9	58.5	47.8
2%		WB	37.1	36.0	43.7	55.4	65.9	71.9	73.9	72.5	70.6	61.3	50.9	40.9	
		MCDB	38.3	38.0	48.9	64.0	75.4	81.8	83.5	81.4	77.2	66.9	53.9	42.8	
5%		WB	33.3	33.7	39.4	49.9	62.6	69.4	71.7	70.6	68.1	58.1	46.8	36.7	
		MCDB	34.6	35.4	44.1	58.7	70.6	77.2	79.9	78.2	73.8	62.8	49.6	38.0	
10%		WB	30.0	31.2	36.6	46.4	59.2	67.0	69.7	68.8	65.1	54.7	43.1	33.8	
		MCDB	31.7	33.6	40.6	53.2	66.8	74.1	77.1	75.3	70.6	59.0	45.7	35.1	
Mean Daily Temperature Range	5% DB	MDBR	18.8	20.8	20.2	21.0	23.3	21.9	22.4	21.8	21.4	17.1	14.4	15.0	
		MCDBR	15.7	18.6	23.5	33.4	32.6	27.8	27.2	26.5	26.5	25.0	19.7	14.2	
		MCWBR	14.8	16.1	16.6	19.8	17.2	13.9	13.1	13.5	15.6	16.6	16.1	13.0	
	5% WB	MCDBR	15.5	16.5	20.7	28.0	25.6	22.9	22.8	22.1	22.3	22.6	18.1	13.6	
		MCWBR	15.1	14.9	15.5	18.6	14.5	12.9	12.3	12.3	13.9	16.1	15.4	12.9	
Clear-Sky Solar Irradiance	taub		0.270	0.287	0.321	0.355	0.372	0.412	0.407	0.394	0.371	0.344	0.310	0.273	
	taud		2.349	2.290	2.338	2.375	2.385	2.293	2.322	2.356	2.414	2.475	2.491	2.393	
	Ebn at Noon		266	283	288	286	284	272	271	271	268	260	250	252	
	Edh at Noon		23	30	34	36	37	41	39	36	31	26	21	20	
All-Sky Solar Radiation	RadAvg		428	730	1164	1447	1637	1804	1789	1563	1215	715	420	310	
	RadStd		61	88	104	123	145	134	149	88	116	74	48	37	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page